

FAIRVIEW AGDM, AB, Canada

WMO: 713470

Lat: 56.0814N

Lon: 118.4394W

Elev: 655

StdP: 93.70

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.1	-29.1	-35.9	0.1	-32.2	-32.6	0.2	-29.0	12.2	2.9	10.7	2.6	1.6	30	0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	27.4	16.1	25.5	15.2	23.8	14.5	17.6	24.5	16.5	23.5	15.6	22.1	3.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	11.5	20.4	13.9	10.7	18.6	12.9	10.0	17.6	52.1	24.8	48.6	23.6	45.8	22.1	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.7	7.6	-36.2	30.4	3.1	1.7	-38.4	31.6	-40.2	32.6	-42.0	33.6	-44.2	34.8		
(5)			-36.5	19.5	3.2	1.7	-38.8	20.7	-40.7	21.7	-42.5	22.7	-44.8	23.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-12.1	-11.5	-6.2	2.7	10.1	14.2	16.0	14.5	9.7	3.4	-6.7	-11.8		
	DBStd	12.21	8.95	8.28	7.68	5.40	4.47	3.07	2.86	3.42	4.34	4.78	7.41	8.15		
	HDD10.0	3518	684	603	504	222	53	3	1	4	56	211	502	675		
	HDD18.3	6010	942	836	762	468	255	129	82	127	258	464	752	933		
	CDD10.0	571	0	0	0	4	57	128	187	142	48	5	0	0		
	CDD18.3	22	0	0	0	0	1	4	11	7	1	0	0	0		
	CDH23.3	477	0	0	0	2	28	81	199	136	32	1	0	0		
	CDH26.7	86	0	0	0	0	4	12	37	30	3	0	0	0		
(14) Wind	WSAvg	3.4	3.0	2.9	3.5	3.9	4.0	3.6	3.3	3.2	3.5	3.5	3.0	2.9		
(15) Precipitation	PrecAvg	458	28	24	19	21	46	75	78	56	39	27	24	25		
	PrecMax	621	61	81	48	64	158	166	160	137	129	67	48	73		
	PrecMin	300	6	4	4	3	6	9	10	10	4	2	0	4		
	PrecStd	81	15	18	12	14	36	42	36	32	26	14	13	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	5.8	9.1	21.6	26.6	28.3	30.0	29.8	26.7	20.6	10.8	5.6		
		MCWB	3.1	2.2	4.6	10.2	13.2	16.1	17.5	16.6	14.8	11.5	5.5	2.6		
	2%	DB	4.4	3.9	6.8	16.2	23.1	25.5	27.4	26.9	23.6	15.5	7.3	3.9		
		MCWB	1.9	1.2	2.8	7.4	11.5	14.6	16.6	16.1	13.8	8.7	3.8	1.4		
	5%	DB	3.0	2.1	5.1	13.2	20.9	23.7	25.5	24.4	20.1	12.6	4.9	1.7		
		MCWB	1.0	0.0	1.8	5.9	10.6	14.0	15.9	15.0	12.4	7.0	2.3	-0.2		
	10%	DB	0.9	0.0	3.6	10.8	18.8	21.7	23.7	22.4	17.6	10.6	2.8	-0.3		
		MCWB	-0.7	-1.9	0.8	4.7	9.6	13.2	15.3	14.4	10.9	5.9	0.6	-1.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	2.7	5.0	10.5	14.4	17.6	19.7	18.3	16.0	12.2	6.1	3.1		
		MCDB	5.4	5.2	8.5	20.7	23.7	25.2	26.0	25.3	24.1	19.7	9.9	5.2		
	2%	WB	2.2	1.4	3.2	8.1	12.4	16.0	18.1	17.1	14.4	9.3	4.1	1.5		
		MCDB	4.1	3.6	6.5	15.7	20.9	23.1	24.4	24.6	21.6	14.4	6.7	3.5		
	5%	WB	1.2	0.2	2.0	6.3	11.2	14.9	17.0	16.0	13.0	7.6	2.5	0.0		
		MCDB	3.0	2.3	4.9	12.3	19.1	21.7	23.7	23.0	19.5	11.5	4.7	1.7		
	10%	WB	-0.7	-1.6	0.9	5.0	10.3	13.8	16.0	15.0	11.6	6.3	0.8	-1.9		
		MCDB	1.1	0.2	3.4	10.3	17.4	20.3	22.4	21.0	16.7	10.0	2.5	-0.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	9.7	10.1	10.2	12.4	11.9	12.7	12.8	11.9	9.6	8.6	8.5		
		MCDBR	8.9	10.4	10.3	14.3	15.6	15.5	16.3	16.4	15.9	13.1	9.6	9.8		
	5% WB	MCWBR	7.0	8.2	7.1	7.7	7.3	7.0	7.2	7.2	7.7	7.5	7.0	8.3		
		MCWBR	8.5	10.1	9.7	13.4	14.3	13.3	14.4	14.6	15.1	12.3	9.1	9.7		
(40) Clear-Sky Solar Irradiance	taub		0.259	0.249	0.296	0.320	0.341	0.350	0.351	0.348	0.308	0.283	0.241	0.223		
	taud		2.177	2.341	2.278	2.387	2.399	2.393	2.406	2.406	2.526	2.548	2.405	2.223		
	Ebn at Noon		646	847	872	903	900	893	883	863	860	803	708	620		
	Edh at Noon		56	70	97	102	108	110	106	100	77	58	44	40		
(44) All-Sky Solar Radiation	RadAvg		0.70	1.66	3.17	4.70	5.58	6.02	5.85	4.76	3.26	1.73	0.77	0.48		
	RadStd		0.07	0.13	0.24	0.26	0.38	0.49	0.35	0.42	0.28	0.20	0.07	0.04		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page